Lab assignments: Logistics

You can use the Lingo software and Excel or some other sowtware for the first one. LINGO is available to download free from www.lindo.com or available in the lab room. A manual and help can be downloaded from the homepage as well as the optimization by Lingo (Shrage)

Leif will be available at scheduled times in the lab room (see schedule for location). It will be one 3 hours of every exercise and one extra, in total 6 occasions. Exercises will be examined and discussed directly at the lab occasion.

These tasks gives in total 3hp with pass or fail as the grade and is performed in groups of 2 or 3 students.

The aim of these assignments is to model different relevant problems using the models we learn in the course. Note also that similar exercises, although simpler, will be on the written exam.

You need to explain the mathematical formulation of the models and give the results orally.

You need to be able to present the model and the results for the teacher at the lab to get a pass.

If you use Lingo the teacher can help you with the required code. Otherwise presenting an accurate general mathematical formulation will help you with the development using other software.

You can use help from friends or AI chats or whatever. Remember that this mainly an exercise to prepare you for the written exam.

1. Inventory management and production

a) Exponentially weighted average

Data are contained on the table below where d (t) is the real demand and u (t-1) the forecast made months before with an initial estimate of 50 units.

	jan	Feb	Mar	Apr	Maj	Jun	Jul	Aug	Sep	Okt	Nov	Dec
d(t)	52	51	60	46	70	50	39	48	60	70	60	
u(t-1)	50*											

- Calculate the forecast u(t) for December and the Mean absolute percentage error (MAPE) for all values of a between 0 and 1. What is the difference in results and what is the meaning of the forecast for different values of a.
- Derive the recursive formula for u(t) based on the original time series in the lecture.

b) ABC Analysis

Make an ABC (Pareto) analysis of the material contained in ABC.xls. Illustrate the results with appropriate diagrams. How many A products is it and what does it mean?

2. Integer programming

Four people at a company can perform six different work tasks and generate the following profit V(i,j).

V(i,j) =(37000, 23000, 20000, 18000, 16000, 6000,
25000, 14000, 14000, 18000, 17000, 8000,
5000, 13000, 10000, 12000, 5000, 5000
10000 3000 13000 7000 2000 16000);

Each works task can only be performed by one person.
They initially have 4, 4, 8 and 8 days respectively to perform all works tasks and all tasks take the same amount of time.
The time it takes them to do each task is 2 days for person 1, 3 days for person 2 and 3, and 4 days for person 4
a) Formulate a general mathematical model and solve the problem for the conditions above using software, for instance, LINGO
b) Change the first person's available total time to 6 days, What conclusion can be drawn?

3. Transshipment model: Probabilistic constraints and integers

A transport network for a product has four suppliers, two transshipment nodes and three customers. The demand for the units in the upcoming month with the three customers is uncertain and therefore historical data has been used to approximate the demand using a normal distribution. The four suppliers (mills) can deliver maximum 400, 350, 300 and 450 units during the period respectively. The cost of transport between the various nodes is shown in tables 2 and 3 below, where A, B, C and D are supply nodes, E and F are transshipment nodes and G, H and I are the customers eg the nodes with normal distributed demand. Answer the questions below and analyze the results.

Table 1: Demanded units estimated as a normal distribution for customers at node G, H and I.

Customer	Mean value	Stdav
G	560	40
H	200	20
I	400	30

Table 2: Cost in SEK/unit from Supply nodes to transshipment nodes

From/to	E	F
A	4	6
B	5	7
C	8	6
D	4	4

Table 3: Cost in SEK/unit from transshipment nodes to demand nodes (customers)

From/to	G	H	I
E	2	3	1
F	2	2	4

a) Depict the transshipment problem as a network and define all sets, parameters and variables in a mathematical way.

b) Formulate the general mathematical model
c) Solve the transshipment model with the mean value as demand. Is there any mill that is not needed in the optimal solution?
d) Extend the model in b. and solve it when the demand is normally distributed according to table 1 with a probabilistic constraint transformed into a deterministic equivalent using a standardized normal distribution. Assume all the probabilistic constraints hold at 90% significance level. Is there any difference to the original solution? Describe this difference. What happens if the significance level is increased?
e) Extend the model when we have fixed cost of operating a mill at A=300, B=200, C=250 and D=225 if it is used and analyze the result. Is there any difference between the solutions in c) and d)? Describe this difference. What happens if the supply at mill 2 increases from 350 to 450 units? And finally what type of model is this?
f) Further extend the model adding transport capacities between the supply nodes and the transshipment node of 350 on every link (Table 2). What is the lower limit of this capacity if it is the same on all these links? Are all the solutions integers? If not solve the integer problem and compare the solutions. Is rounding, ok?

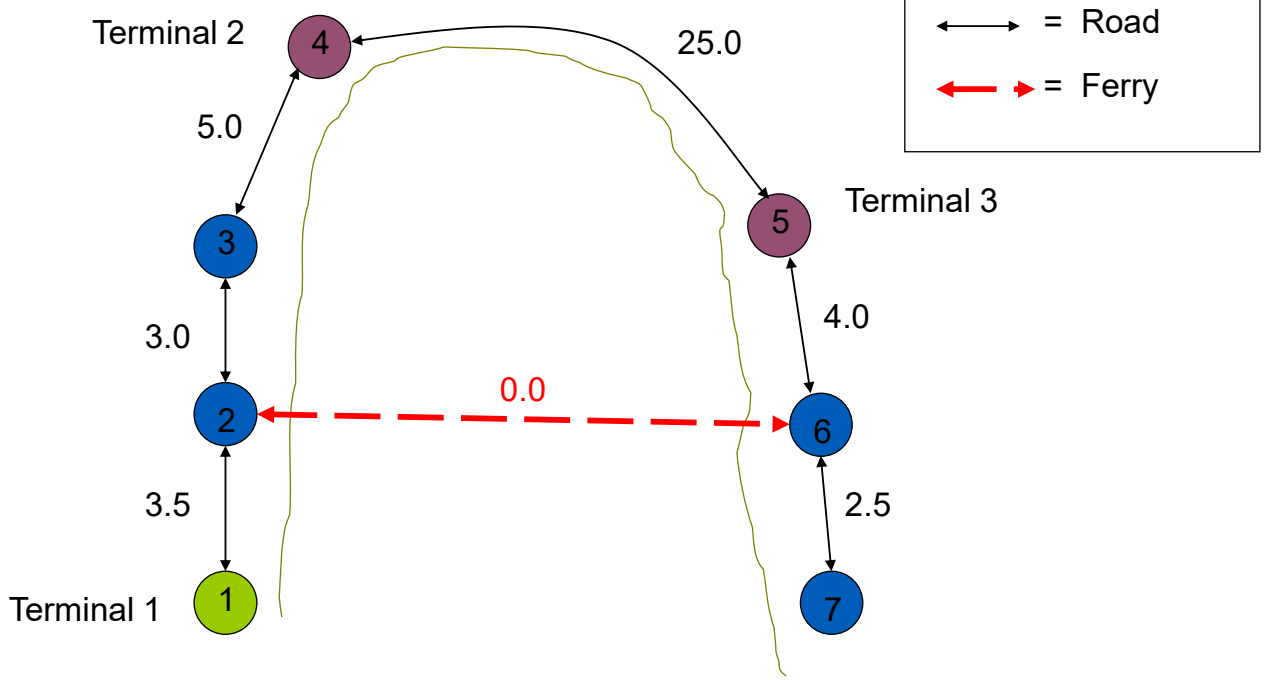
4. Intermodal transport planning

Trucks transporting goods today on roads around a bay and we want to analyze how much the truck traffic on the road's decreases in Swedish mil (1 mil =10 km) in a year when one or more ferries are used to transport trucks over the bay. We are interested in environmental benefits of such a procedure, ie how much ferries will reduce the actual trucks transport in the area.

The number of trucks with destinations is listed below.

Number of trucks

Terminal	1	2	3	4	5	6	7	Total
1	0	900	750	40	10	600	550	2850
4 (2)	100	2000	1100	0	150	1400	1250	6000
5 (3)	110	4000	2200	200	0	3300	2440	12250



The trucks are currently located at the terminals in city No. 1, 4 and 5 in the table above example there are 2850 cars at the terminal in city 1, etc.

We want to minimize the distance that we use trucks on the roads. Therefore, the route by ferry is set to 0 in the figure above. The distances between cities are indicated in the figure above. For example, it is 3 Swedish miles (30km) distance to drive between city 2 and 3.

a. Formulate a general mathematical model based on the network above

b. Minimize the truck traffic using the specified data given that one ferry can take 2000 trucks during the year **in either direction**.

c. What happens if the number of ferries increases. How many ferries are the maximum required? Illustrate this with a diagram in for instance Excel.

d. Simulate what happens if the total capacity is decreased on the road between 2 and 3 (and 3 and 2) in the first case of the ferry capacity of 2000 in each direction. What capacity is needed for no effect on the total solution?

5. Empty freight wagons

In a small railway network there are 5 stations, numbered 1 to 5. The stations have s(1),…,s(5) empty freight cars today and need d(1),…,d(5) empty freight cars tomorrow. During the day and the coming night, empty freight cars will be sent between the stations to meet tomorrow's needs. The total need for empty cars is equal to the total supply. Trains can be formed in all relationships between the stations since there is a railway between all stations. There is no limit to how long the trains can be. Sending a car the distance (i,j) costs c(i,j) per car. The locomotive and personnel cost to run a train the distance (i,j) is g(i,j). Congestion in the timetable means that if cars are sent from stations 1 to 4 and 5, the cost P is added.

a) Draw the network model, which model type is this?

b) Formulate the general mathematical problem of minimizing the cost of meeting the empty wagon demand.

c) Below are the data for the problem. Use this data to solve the problem formulated in b)

d) What will be the result with and without the congestion condition?

e) Simulate at what cost P the freight wagons from 1 to 4 and 5 still will be transported?

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s= 20 5 0 6 4;

d= 5 3 2 13 12;

P=50;

c = 0 10 12 15 18

10 0 5 12 18

12 5 0 4 12

15 12 4 0 4

18 18 12 4 0;

g=0 25 29 35 50

25 0 13 29 48

29 13 0 10 27

35 29 10 0 9

50 48 27 9 0;